

Sequences & Patterns



Arithmetic sequences

- 1 Find the 10th term of the sequence 5, 8, 11, 14, ...
 - 2 Find the 15th term of the sequence 7, 12, 17, 22, ...
 - 3 An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.
 - 4 The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.
 - 5 An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?
 - 6 The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.
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Geometric sequences

- 7 Find the next term of the sequence 2, 4, 8, 16, ...
 - 8 Find the 6th term of the sequence 3, 6, 12, 24, ...
 - 9 A geometric sequence has first term 5 and common ratio 3. Find the 5th term.
 - 10 The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.
 - 11 A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.
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Identifying the type of sequence

- 12 Is 5, 10, 20, 40, ... arithmetic, geometric, or neither?
 - 13 Is 6, 10, 14, 18, ... arithmetic, geometric, or neither?
 - 14 Is 1, 4, 9, 16, ... arithmetic, geometric, or neither?
 - 15 Is 100, 90, 81, 72.9, ... arithmetic, geometric, or neither?
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Finding missing terms

- 16 Find the missing term: 3, __, 11, 15 (arithmetic).
- 17 Find the missing term: 4, __, 36 (geometric, all terms positive).
- 18 Insert two arithmetic means between 5 and 20, forming the four-term sequence 5, __, __, 20.
- 19 The sequence x , $2x + 1$, $4x - 1$ is arithmetic. Find x .
- 20 Find the two missing terms of the geometric sequence 2, __, __, 54.
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Sum of an arithmetic series

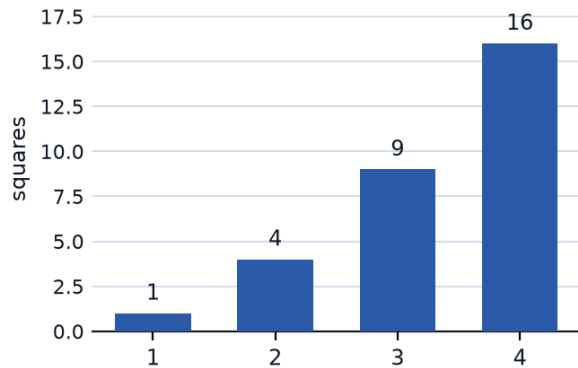
- 21 Find the sum of the first 10 terms of the sequence 2, 5, 8, 11, ...
- 22 Find the sum of the first 20 positive integers.
- 23 Find the sum of the first 12 terms of the sequence 10, 7, 4, 1, ...
- 24 An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.
- 25 Find the sum of the first 25 positive even integers (2, 4, ..., 50).
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Special sequences

- 26 The sequence 1, 4, 9, 16, 25, ... consists of perfect squares. Find the 8th term.
- 27 Find the 6th triangular number, from the sequence 1, 3, 6, 10, 15, ...
- 28 A sequence starts 2, 3, 5, 8, 13, ..., where each term is the sum of the two terms before it. Find the next two terms.
- 29 Find the 7th term of the sequence of powers of 2: 1, 2, 4, 8, 16, ...
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Visual and growing patterns

- 30** The bar chart shows the number of small squares used to build patterns 1 through 4. Predict the number of squares in pattern 6, and state the rule.



- 31** A pattern of dots grows as 3, 7, 11, 15, ... (each stage adds 4 dots). Find the number of dots in stage 10.
- 32** A row of matchstick triangles uses 3, 5, 7, 9, ... matchsticks for 1, 2, 3, 4 triangles. Find the number of matchsticks for 15 triangles in a row.
- 33** A row of matchstick hexagons uses 6, 11, 16, 21, ... matchsticks for 1, 2, 3, 4 hexagons. Find the matchsticks needed for 9 hexagons in a row.

Word problems

- 34** A theater has 20 seats in the first row, and each following row has 3 more seats than the row before it. How many seats are in the 12th row?
- 35** A ball is dropped and after each bounce rises to half its previous height. If it first rises to 80 m after being dropped, how high does it rise after the 4th bounce?
- 36** A savings plan deposits 100 riyals in week 1, then increases the deposit by 20 riyals each following week. Find the total saved after 8 weeks.
- 37** A bacteria culture starts with 50 bacteria and doubles every hour. How many bacteria are there after 6 hours?

More on series and sequences

- 38** Find the sum of the first 8 terms of the geometric sequence 1, 2, 4, 8, ...
- 39** A geometric sequence has first term 4 and common ratio 2. Find the sum of its first 6 terms.

- 40 An arithmetic sequence has 3rd term 11 and 7th term 27. Find the common difference and the first term.
- 41 The n th term of a sequence is $T_n = 3n - 2$. Find the 12th term, and state whether the sequence is arithmetic.
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Pattern reasoning

- 42 A sequence of tile patterns is built so that stage n uses $n^2 + 1$ tiles. How many tiles are needed for stage 7?
- 43 The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in an octagon ($n = 8$).
- 44 A sequence follows the rule "double and add 1", starting at 3 as the 1st term. Find the 4th term.
- 45 The number of handshakes when n people each shake hands with every other person once is $\frac{n(n-1)}{2}$. Find the number of handshakes among 10 people.